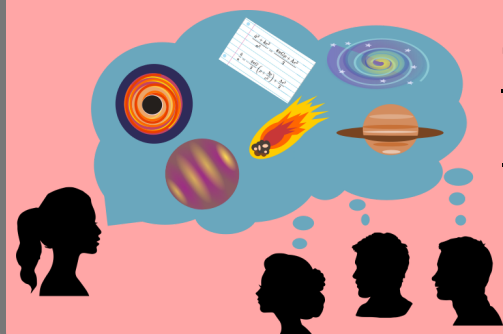




Telling Stories of the Stars: Science Communication for Astronomers

Spring 2025: Tuesdays & Thursdays 10:10am-11:25am

Instructor: Ryan Golant (feel free to call me Ryan -- no title or honorific preferred)

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Phone: 703-582-7957

Twitter: @rehg98

Office hours: Formally, Tues/Thurs 11:30am-12:30pm (i.e., right after class). However, I'm happy to meet with any of you anywhere at any time 😊 (given advanced notice)

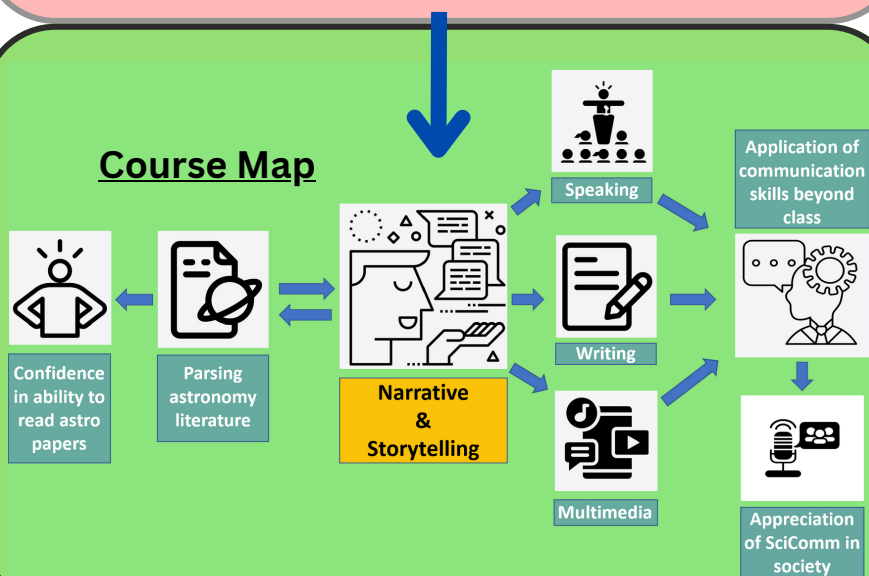
Course Learning Objectives

1. Students should be able to **identify the key components of a published astronomy paper** and **translate the paper's ideas** into a narrative structure.
2. Students should **feel confident in their abilities to critically read and discuss current astronomy literature** with their peers.
3. Students should be able to **communicate astronomy concepts and research results** clearly and engagingly through a variety of media and to a variety of audiences.
4. Students should be able to **apply critical reading, writing, and speaking skills** in their future careers and everyday lives. →
5. Students should be able to **express the importance of clear and accurate science communication** in modern society.

Course Description

All humans, all cultures, worldwide, are united by the night sky – we all share the same moon, the same stars, the same Milky Way. Astronomers – as interpreters of that which lies just out of reach yet binds us all together – have unique responsibilities not only as researchers, but also as communicators; **to not share our findings with the world would be an injustice**. How, then, can we best **communicate our knowledge as astronomers**? In this seminar course for upper-level undergraduate astronomy majors and early-year graduate students, we will explore how storytelling and science communication go hand-in-hand. In this course, we will practice **deconstructing and interpreting astronomy literature** via the use of narrative, a skill invaluable to anyone pursuing astronomy research. We will **communicate astronomy-related concepts and research results clearly and engagingly** through a variety of narrative media, including writing, public speaking, video production, podcasting, and more; we will learn to **adapt our communication methods based on context and audience**, ranging from the public to fellow astronomers. By the end of this course, we will be fluent in a range of communication techniques and will possess a deeper awareness of which techniques best suit our individual personalities and abilities. These communication techniques will be crucial in any career context, within or outside of academia.

Course Map



Teaching Philosophy

My primary goal in teaching this course is to help you develop skills that'll be useful throughout your career, in academia or otherwise. I am here to **help you learn**. In turn, I also hope to **learn from you** -- I myself am still growing as an astronomer and as a science communicator. I thus aim to create a learning environment that's safe and accessible to *everyone* so that we may all equally contribute to the discussion. I wish to minimize the inherent power differential between student and teacher so that we can interact as peers. Ultimately, this course will be shaped and driven by *you*, the students; please don't hesitate to provide feedback throughout the semester, whether it's about course content, course policies, class structure, or my own teaching. **This syllabus is a living document and will change based on your input.**

I expect you to..

- **Come to class on time and actively participate in class to the best of your abilities.**
- **Complete all assigned work -- including any necessary pre-class reading or writing -- in a timely manner.**
- **Respect the ideas and contributions of your peers.**
- **Be transparent with me about any issues that may affect your ability to perform optimally in this class.**
- **Provide regular feedback on any and all aspects of the course so that we may work towards establishing an optimal learning environment.**

You should expect me to...

- **Come to class at least 5 minutes early and be fully prepared for every class session.**
- **Assign no more than 5 hours of work outside of class each week.**
- **Grade and provide feedback on all submitted assignments in a timely manner**
- **Facilitate stimulating full-class discussions and worthwhile small-group activities to the best of my abilities.**
- **Respect the ideas and contributions of all students.**
- **Adapt my teaching methods to suit the needs of the greatest number of students.**
- **Respond meaningfully and promptly to any and all student input.**

Prerequisites: Students should have a basic survey-level knowledge of modern astronomy -- enough background knowledge to be able to read and understand (at surface level) a published astronomy research paper. I will try my best to make sure no one is falling behind in class due to lack of content knowledge; the focus of this class is on communication skills, not on conceptual astronomy knowledge. I am happy to recommend additional resources for background reading upon request.

Technology Policy: The use of technology is permitted in this class to the extent that it aids in your learning and does not significantly distract others. Many in-class activities will require a laptop, a tablet, or a similar device with access to the internet; if this poses a problem for you, please let me know. The astronomy department has spare Macbooks for use if you aren't able to bring a laptop to class; please let me know before class if you'll be needing a laptop for that day.



Participation

In astronomy, communication does not just mean talking *at* people, but also talking *with* people; at conferences, within collaborations, or simply with your advisor, asking questions and engaging in discussions is necessary to acquire knowledge, to form connections with other astronomers, and to advance the field as a whole. As such, **participation – emulating what’s required in a professional astronomer’s career – is a critical part of this course** (see [Course Learning Objective #2](#)).

Within this course, “participation” will include coming to class prepared for discussion, asking questions (to both me and your peers), answering questions (from both me and your peers), and engaging in peer review and other in-class activities; I expect everyone to engage in these tasks to the best of their abilities, and I will try my best to provide accommodations to facilitate participation (e.g., by providing access to a semi-anonymized discussion forum if you don’t wish to ask a question in front of your peers). Participation in class will allow us (including me!) to learn from each other; I plan to participate in discussions with you all, since I myself still have much to learn. **This peer-to-peer learning will work best if we maximize the number of people involved.**

Participation will be evaluated based on a single-point rubric (below). At the start of the course, you will devise a set of goals for class participation, and we will subsequently meet one-on-one three times throughout the semester to discuss your level of participation and to revise your goals moving forward; at each of these meetings, we will discuss the extent to which you met your goals for that portion of the class and, **together, we will assign a participation grade accordingly.** You will receive three participation grades throughout the semester (one for each meeting); these grades will be weighted progressively – 10% of the total participation grade for the first grade, 30% for the second, and 60% for the third – to emphasize growth throughout the course.

Participation Rubric

Concerns (Areas that Need Improvement)	Criteria (Standards for Performance)	Advanced (Evidence of Meeting or Exceeding Standards)
	Class Preparation The student comes to class having completed the requisite preparatory work (e.g., finding articles, reading articles, reading peer work, etc.) and having thought of questions or other insights for driving in-class discussion.	
	Questions/Discussions The student engages in class discussions by asking or answering questions, providing ideas or examples, or challenging the ideas of others (both peers and the instructor). These actions may be executed verbally or via our (semi-)anonymized in-class chat/forum moderated by the instructor.	
	Peer Review The student makes their best effort to provide constructive, polite criticism on work produced by their classmates. The student both identifies aspects of their own work on which they’d like targeted feedback and addresses targeted feedback areas in their peers’ work.	

Assignments

Since science communication can take on many forms, the assignments in this class will vary widely. In the first half of the course, we will mostly be focusing on **communicating science to fellow astronomers**; this will involve both *deconstructing* published astronomy papers to extract their key points and results ([Course Learning Objectives #1 and #2](#)) and *reconstructing* papers as we learn to write and speak about our own results ([Course Learning Objective #3](#)). In the second half of the course, we will be experimenting with and **exploring various ways of communicating astronomy to audiences outside our field**, including scientists outside of astronomy, younger students and children, and the broader public ([Course Learning Objectives #3 and #4](#)). In general, assignments will fall into three categories:

Small-scale skill-builders: This class will be largely interactive and hands-on. Aside from a small amount of lecturing from the instructor, some full-class discussions on readings done outside of class, and peer review of your classmates' work, much of class time will be spent working collaboratively on short-term assignments or mini-projects (e.g., critiquing a piece of media, storyboarding a scientific paper, preparing a small academic poster, etc.); these small-scale assignments are designed to help you build a wide array of skills pertinent to science communication and beyond ([Course Learning Objectives #1, #3, and #4](#)). As appropriate, these mini-projects will also be accompanied by small group-based presentations to the class and/or individual reflections. **You will receive feedback on these assignments both from your peers and from me, and you will receive a grade from me** (based on the assignment, this may just be a completion grade).

Weekly reflections: At the end of each week (by midnight on Friday), you will submit a **brief (~300-word) reflection** summarizing what you learned the past week, which skills you've developed, which skills you'd still like to develop, and how these skills could be applied outside of class ([Course Learning Objective #4](#)). This reflection should also include a 1-2 paragraph critique of an academic paper, a science news article, or a popular science piece that you encountered that week, briefly summarizing the main ideas of the work, what elements aided in the communication of the main ideas, and how the work could've communicated its ideas more effectively ([Course Learning Objective #5](#)); we will discuss some of these works in class the following week. For these reflections, you may submit either a written document or a spoken recording, whichever is easier. **You will receive a grade and feedback on these reflections from me.**

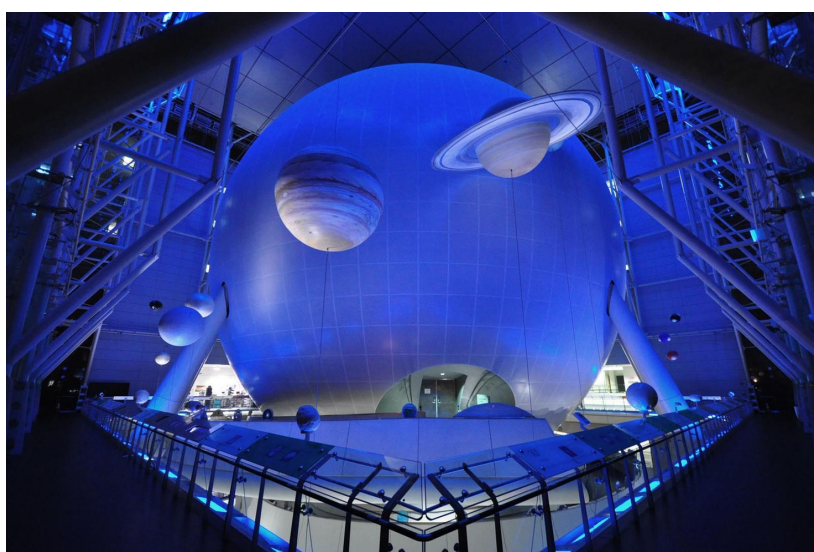
Long-term projects: There will be **two long-term projects in this course**. In the first half of the course, you will be preparing a brief astronomy talk or poster presentation for a small in-class astronomy "conference" ([Course Learning Objectives #1 and #2](#)). In the second half of the course, you will be crafting a piece of popular science media in a format of your choosing ([Course Learning Objective #3](#)). These long-term assignments will be broken down into many smaller pieces, to be completed over multiple weeks; **these sub-assignments will be graded purely for completion, but will be peer-reviewed in class** (I am also open to giving feedback on these sub-assignments upon request). **For the final product, you will receive feedback both from your peers and from me, and you will receive a final grade from me.** More details on these long-term assignments can be found below.

(rubrics for all assignments can be found under "Files/Rubrics" on CourseWorks)

Grade Breakdown

Assignment	Point value
Skill-builders	5
Weekly reflections	10
Participation	15
Conference talk/poster	30
Popular science piece	40
TOTAL	100 points

Extra credit: There may be some opportunities for extra credit throughout the semester involving active engagement with astronomy communication -- for instance, visiting a science museum (like the American Museum of Natural History) and writing a critique of an astronomy display, or attending and reflecting on a public astronomy talk through Columbia Astro's public outreach program or through the Amateur Astronomer's Association. These extra credit opportunities will be announced as they arise (at least two weeks in advance); the amount of extra credit possible will differ based on the activity.



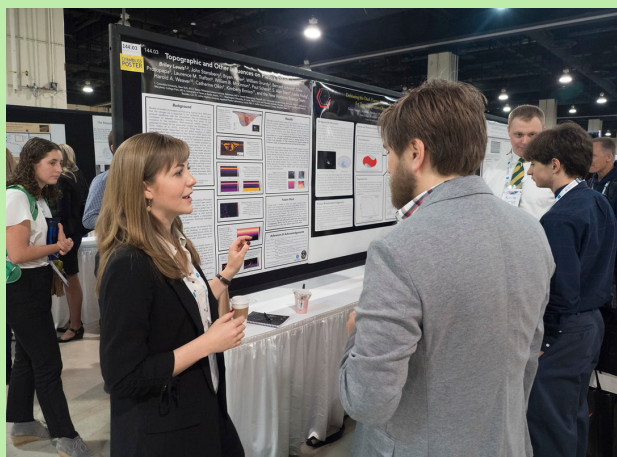
Long-Term Projects

Midterm Project: Astro mini-conference

You've been invited to share your work at a prestigious astronomy conference! For this assignment, you'll choose between delivering a 10-minute academic conference talk to the class or designing and briefly presenting an academic poster. The twist: you'll be presenting on a paper that *someone else has written*. This assignment will require you to apply the paper-reading skills you've developed in the first half of the course to effectively parse and communicate recent astronomy scholarship. You'll be completing this assignment in small chunks over the course of six weeks, with checkpoints for completion along the way (see course calendar for a timeline). Your tasks will be to:

1. **Pick a paper** that's been posted on astro-ph (<https://arxiv.org/archive/astro-ph>) in the last 3 months. All papers must be approved by me to make sure they're of reasonable length and accessibility.
2. Read the paper in its entirety and **submit a brief (~250-word) summary** of the paper's key ideas.
3. **Select the graphics** (both from your chosen paper and from related papers or media) that best tell the paper's story; you will submit these graphics alongside 1-paragraph descriptions (in your own words) for each image.
4. **Construct an annotated "storyline"** for the paper, connecting your chosen graphics with a logical flow; for each of the key components of the paper (i.e., the introduction, the methods, the results, and the conclusions), you will place a 1-paragraph summary on the storyline.
5. Flesh out this storyline and **develop it into a written script** for your talk or poster presentation.
6. **Develop visuals** (i.e., slides or a poster) to accompany your script.
7. **Participate in the mini-conference** as both a presenter and an active listener!
8. Submit a **300-word post-project reflection** summarizing what you learned from the assignment, what skills you developed (and how you plan to apply these skills in the future), what you enjoyed about the assignment, and what you found challenging about the assignment

I will be providing direct feedback on your submissions for steps 1, 2, 7, and 8; for steps 3, 4, 5, and 6, you will be receiving peer feedback in class (though I am happy to review materials upon request). This project will total **30 points of your overall course grade**: 5 points for completing each of the checkpoint assignments on time, 15 points for the conference presentation itself, and 10 points for the reflection; you can find the presentation and reflection rubrics under "Files" on CourseWorks.



Final Project: Popular astronomy piece

You've just gotten the scoop on the latest astronomy breakthrough and you have to let the world know about it! For this final assignment, you will be using a medium (or media) of your choice to fully communicate the context and content of a recent astronomy paper to the public. This assignment will thus engage all the knowledge and skills you have acquired throughout this course, from dissecting a scientific paper to identifying the key qualities of effective science communication to analyzing the strengths and weaknesses of various communicative media. You will be completing this assignment in small chunks over the course of six weeks, with checkpoints for completion along the way (see course calendar for a timeline). Your tasks will be to:

1. **Pick a paper** that's been posted on astro-ph (<https://arxiv.org/archive/astro-ph>) in the last 3 months; you may not choose the paper that you used for your midterm project. All papers must be approved by me to make sure they're of reasonable length and accessibility.
2. Read the paper in its entirety and **submit a brief (~250-word) summary** of the paper's key ideas.
3. **Choose a medium** (or multiple media) to communicate your paper's key ideas and submit a ~100-word justification for why this medium is a good choice for this particular piece.
4. **Craft a short, eye-catching pitch** (either written or recorded) summarizing the key elements of your piece in a way that's exciting to a general reader or viewer.
5. After submitting your pitch, you will build up your piece component-by-component, first describing the **context**, then summarizing the **methods and results**, and then wrapping up with the **conclusions and key takeaways**; you will tie all these individual components together into a logically flowing narrative, complete with an interesting hook at the beginning of the piece and a creative send-off at the end of the piece.
6. **Share your piece with the class** during our final class period, which will be dedicated to informal, ungraded "show-and-tell."
7. Submit your **final piece** alongside a **final, revised pitch**.
8. Submit a **300-word post-project reflection** summarizing what you learned from the assignment, what skills you developed (and how you plan to apply these skills in the future), what you enjoyed about the assignment, and what you found challenging about the assignment
9. (Optional, but encouraged) **Submit your piece to a professional science news outlet** or upload your piece somewhere online!

I will be providing direct feedback on your submissions for steps 1, 2, 4, 7, and 8; for steps 3, 4, 5, and 6, you will be receiving peer feedback in class (though I am happy to review materials upon request). This project will total **40 points of your overall course grade**: 5 points for completing each of the checkpoint assignments on time, 10 points for the pitch, 15 points for the popular astronomy piece itself, and 10 points for the reflection; you can find the pitch, piece, and reflection rubrics under "Files" on CourseWorks.



Course Calendar

Note: Assignments in red are due by the start of class, while assignments in blue are due by midnight on Friday evening. All readings should be completed before class.

	Tuesday	Thursday
Week 1 (Jan 16)	<u>Topic:</u> What <i>is</i> science communication? What is this class? <u>Reading:</u> Class syllabus! <u>Due:</u> Syllabus annotation (see CourseWorks announcement)	<u>Topic:</u> How do we make SciComm inclusive and accessible? <u>Reading:</u> 1) <u>Inclusive Science Communication Starter Kit</u>; 2) <u>"Science Communication Demands a Critical Approach That Centers Inclusion, Equity, and Intersectionality"</u> (Katherine N. Canfield et al. 2020) <u>Due:</u> N/A
Week 2 (Jan 23)	<u>Topic:</u> How do you read an astronomy paper? <u>Reading:</u> 1) <u>"How to (seriously) read a scientific paper"</u> (Elisabeth Pain 2016); 2) <u>"Journal Articles in Astronomy"</u> (Dan Gifford 2011); 3) Three-part Astrobites series on <u>"Tools for Reading Papers"</u> (Michael Hammer 2017-2018) <u>Due:</u> N/A	<u>Topic:</u> How do scientific papers tell a story? (Part 1) <u>Reading:</u> 1) <u>"Storyboarding for Biology: An Authentic STEAM Experience"</u> (Caryn Babaian & Sudhir Kumar 2022); 2) Choose one of the following three papers: <u>"A Jupiter-mass companion to a solar-type star"</u> (Michel Mayor & Didier Queloz 1995); <u>"A Bright Millisecond Radio Burst of Extragalactic Origin"</u> (Duncan R. Lorimer et al. 2007); <u>"The Redshift of Extragalactic Nebulae"</u> (Fritz Zwicky 1933) <u>Due:</u> N/A
Week 3 (Jan 30)	<u>Topic:</u> How do scientific papers tell a story? (Part 2) <u>Reading:</u> N/A <u>Due:</u> Finish storyboarding mini-assignment	<u>Topic:</u> How do you write an astronomy paper? <u>Reading:</u> <u>"Writing Scientific Papers in Astronomy"</u> (Johan H. Knapen et al. 2021) <u>Due:</u> Pick a paper for astro mini-conference project; submit brief summary of paper

	Tuesday	Thursday
Week 4 (Feb 6)	<u>Topic:</u> How do you write a good astronomy paper? <u>Reading:</u> 1) "<u>Writing a Clear and Engaging Paper for All Astronomers</u>" (Leslie Sage 2003); 2) "<u>On Writing Well</u>" pages 3-46 (William Zinsser 1976) <u>Due:</u> N/A	<u>Topic:</u> How do you edit an astronomy paper? <u>Reading:</u> 1) "<u>Levels of editing of a scientific paper</u>" (Dawn Field 2018); 2) "<u>On Writing Well</u>" pages 68-92 and 148-165 (William Zinsser 1976) <u>Due:</u> Submit annotated graphics for mini-conference project; Finish paper writing mini-assignment
Week 5 (Feb 13)	<u>Topic:</u> How do you give an astronomy conference talk? <u>Reading:</u> 1) "<u>Speak your science: How to give a better conference talk</u>" (Emily Lakdawalla 2018), plus the <u>accompanying talk</u>; 2) "<u>The Cognitive Style of Powerpoint</u>" (Edward Tufte 2003) <u>Due:</u> N/A	<u>Topic:</u> How do you speak publicly about your astronomy research? <u>Reading:</u> 1) "<u>How to give a great scientific talk</u>" (Nic Fleming 2018); 2) "<u>The Remarkable Vision of JWST</u>" (David Helfand 2023) <u>Due:</u> Submit annotated storyline for mini-conference project
Week 6 (Feb 20)	<u>Topic:</u> How do you make a conference poster? (Part 1) <u>Reading:</u> 1) "<u>How to Make an Award-Winning Scientific Paper</u>" (Meredith Rawls 2016); 2) "<u>How to prepare a scientific poster</u>" (Elisabeth Pain 2022); 3) "<u>Designing conference posters</u>" (Colin Purrington 2019) <u>Due:</u> N/A	<u>Topic:</u> How do you make a conference poster? (Part 2) <u>Reading:</u> N/A <u>Due:</u> Submit presentation script for mini-conference project; Finish poster mini-assignment
Week 7 (Feb 27)	<u>Topic:</u> Guest discussion with Prof. David Helfand: How do you effectively communicate astronomy? <u>Reading:</u> Chapter 1 of "<u>The Universal Timekeeper: Reconstructing History Atom by Atom</u>" (David Helfand 2023) <u>Due:</u> N/A	<u>Topic:</u> [Working session for astro mini-conference project] <u>Reading:</u> N/A <u>Due:</u> N/A

	Tuesday	Thursday
Week 8 (March 6)	<u>Topic:</u> Astronomy mini-conference presentations!!! (Day 1) <u>Reading:</u> N/A <u>Due:</u> Finish mini-conference presentation (if presenting today)	<u>Topic:</u> Astronomy mini-conference presentations!!! (Day 2) <u>Reading:</u> N/A <u>Due:</u> Finish mini-conference presentation (if presenting today); Submit finalized presentation script + slides/poster
Week 9 (March 13)	<u>Topic:</u> How do you write about astronomy for the public? <u>Reading:</u> 1) "On Writing Well" pages 265-285 (William Zinsser 1976); 2) "Record-Breaking Supernova Is Part of a New Class of Objects" (Briley Lewis 2023); 3) "An Intracuster Investigation via Radio Rotation" (Ryan Golant 2022) <u>Due:</u> N/A	<u>Topic:</u> How do you get an astronomy news story published? <u>Reading:</u> 1) "What Makes an Astronomy Story Newsworthy?" (Rick Fienberg 2019); 2) "How to pitch to a science editor" (SciDev.Net 2015); 3) Read three pitches from <u>The Open Notebook's pitch database</u> <u>Due:</u> Submit reflection for mini-conference project; Finish pitch mini-assignment
Week 10 (March 20)	<u>Topic:</u> Spring break!! <u>Reading:</u> N/A <u>Due:</u> Have fun.	<u>Topic:</u> Spring break!! <u>Reading:</u> N/A <u>Due:</u> Have more fun.
Week 11 (March 27)	<u>Topic:</u> How do we approach controversial topics in SciComm? <u>Reading:</u> 1) "Editing Controversial Science" (Laura Helmuth 2022); 2) "Misinformation" (Yasmin Tayag 2022) <u>Due:</u> N/A	<u>Topic:</u> Guest discussion with Prof. Rebecca Oppenheimer: What does astronomy journalism look like in practice? <u>Reading:</u> 1) "Missing Memories of the Universe" (Rebecca Oppenheimer 2020); 2) "Making sense of the exoplanet zoo" (Rebecca Oppenheimer 2016); 3) "Hollywood Can Take On Science Denial" (Rebecca Oppenheimer 2022) <u>Due:</u> Pick a paper for popular astronomy project; submit brief summary of paper

	Tuesday	Thursday
Week 12 (April 3)	<u>Topic</u>: How do you produce popular astronomy videos? <u>Reading</u>: 1) "<u>A Journey to the End of the Universe</u>" (David Kipping 2020); 2) "<u>JWST just made the 'Crisis in Cosmology' WORSE</u>" (Becky Smethurst 2022); 3) Watch a few TikToks by <u>astro_alexandra</u> and <u>astrokirsten</u> <u>Due</u>: Choose a medium for your popular astronomy project and submit a brief justification of your chosen medium	<u>Topic</u>: Guest discussion with Prof. David Kipping: How do you maintain a popular astronomy YouTube channel? <u>Reading</u>: 1) "<u>Watching the End of the World</u>" (David Kipping 2020); 2) "<u>Why we might be alone in the Universe</u>" (David Kipping 2019) <u>Due</u>: Submit draft of pitch for popular astronomy project
Week 13 (April 10)	<u>Topic</u>: Guest discussion with Dr. Moiya McTier: How do you make a career out of astronomy communication? <u>Reading</u>: Chapter 1 of "The Milky Way: An Autobiography of Our Galaxy" (Moiya McTier 2022) <u>Due</u>: N/A	<u>Topic</u>: How is astronomy communication connected to astronomy policy? <u>Reading</u>: 1) "<u>Let's talk about policy, baby</u>" (Mia de los Reyes 2017); 2) Astrobit.es' "<u>Guide to science policy</u>"; 3) "<u>SpaceX and astronomers come to agreement on reducing Starlink astronomy impact</u>" (John Loeffler 2023) <u>Due</u>: Submit draft of context/background for popular astronomy project
Week 14 (April 17)	<u>Topic</u>: What else do you want to learn about astronomy communication? (Part 1) <u>Reading</u>: [based on class interest poll] <u>Due</u>: Submit draft of methods and results for popular astronomy project	<u>Topic</u>: What else do you want to learn about astronomy communication? (Part 2) <u>Reading</u>: [based on class interest poll] <u>Due</u>: Submit draft of conclusions and key takeaways for popular astronomy project
Week 15 (April 24)	<u>Topic</u>: [Working session for popular astronomy project] <u>Reading</u>: N/A <u>Due</u>: N/A	<u>Topic</u>: Popular astronomy project show-and-tell!!! <u>Reading</u>: N/A <u>Due</u>: Finish popular astronomy piece; submit final piece + pitch; submit project reflection by May 5

Additional Resources

- **Astrobetter**: A blog covering anything related to professional development in astronomy, including science communication.
- **Astrobites**: While primarily a written astro-ph digest aimed at undergraduate astronomers, Astrobites is a fantastic resource for anything adjacent to astronomy communication (and beyond); see Astrobites' "Beyond" posts and "Guides" for tons of helpful advice on navigating an early career in astronomy.
- **Astro-ph**: An archive of open-access astronomy papers, updated daily (except on weekends); a convenient resource for keeping up-to-date with astronomy literature (but be warned that not all the papers on astro-ph have been peer reviewed).
- **Astrophysics Data System (ADS)**: A powerful, comprehensive, and easy-to-use database of published astronomy papers.
- **Columbia Astronomy Public Outreach**: The Columbia astronomy department offers many opportunities to both experience public astronomy communication in action and to get involved with astro communication yourself.
- **The Inclusive SciComm Symposium**: A wide array of helpful resources for promoting equity, inclusivity, and accessibility in science communication.
- **The Open Notebook**: A wealth of articles, online courses, and other training opportunities for aspiring science journalists.
- **The SciCommers Network**: A community of scientists and engineers interested in engaging the public through science communication; includes a community Slack channel, opportunities for peer feedback on science writing, and regular Zoom calls with SciComm experts.

... and, of course, anyone in the **Columbia astronomy department**! Everyone in the department is friendly and more than happy to talk about anything astronomy-related, so always feel free to reach out to a department member directly or to ask me to connect you.

**If I've missed any useful astronomy
communication resources, please let me know
and I'll add them to the list!**

